SIMATS ENGINEERING
ASSESSMENT TOOL 3

NAME: RAJANEDI DATI PRAVALLIKA **REG.NO:** 192525473
COURSE NAME: CSA1603 **COURSE CODE:** Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:50**

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items *(2)*	Comparison Depth(2.5)	Use of Evidence *(2)*	Critical Thinking *(2)*	Clarity of Presentation *(1)*	***Total(10)***
Weightage	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **S.Blessika (192525251)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.Blessika

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

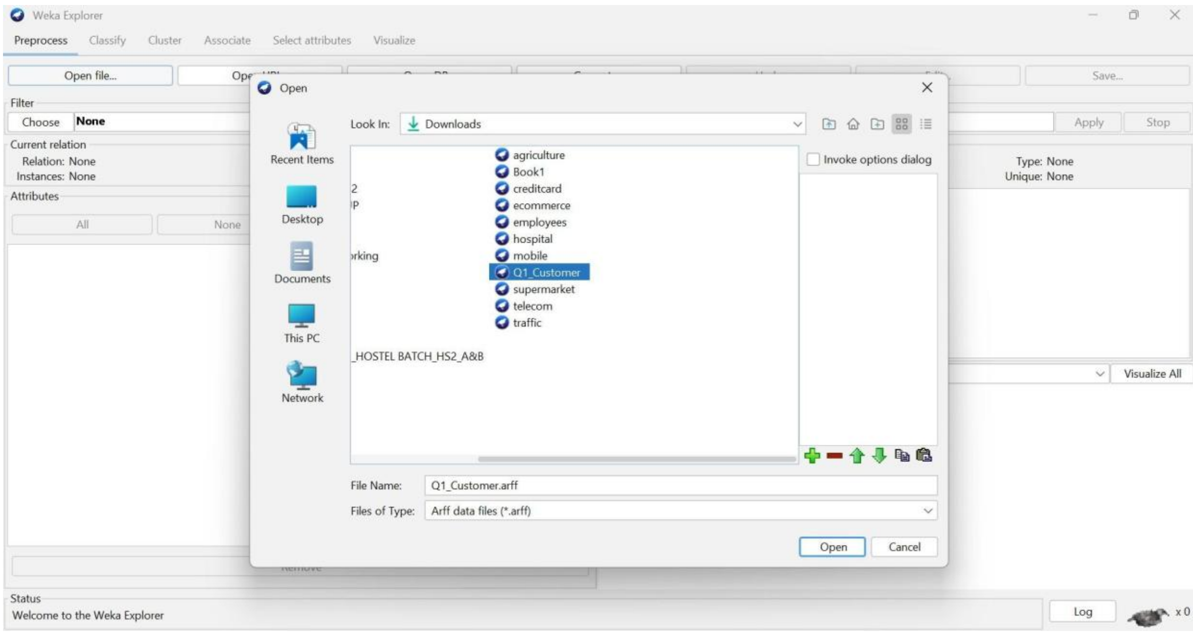
Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

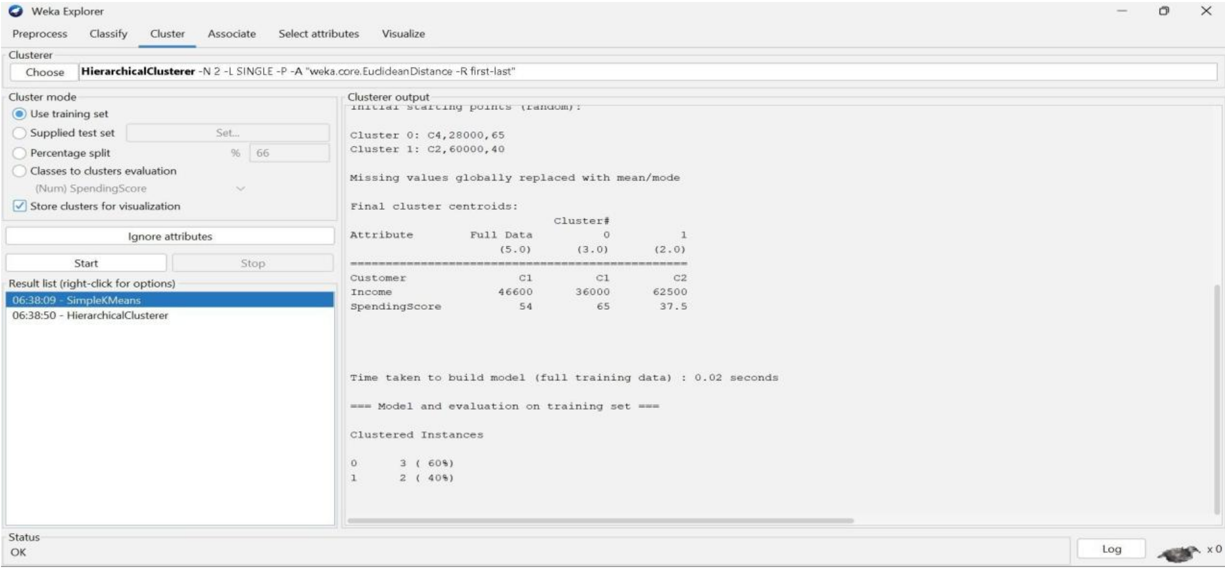
Time duration :45 Minutes
Assessment Tool 3 (Low)

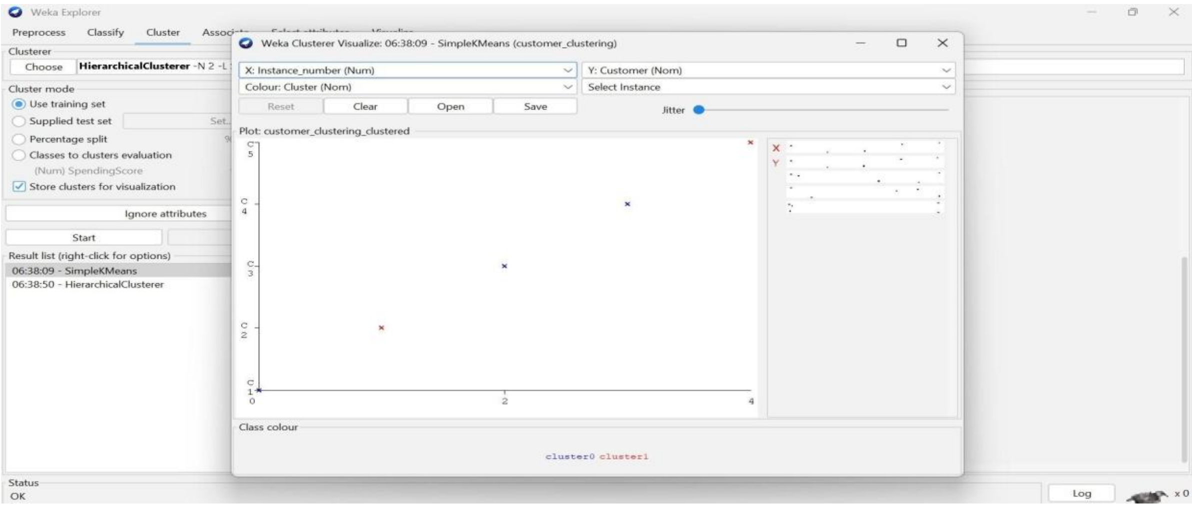
Q1 . Dataset: Customer Data

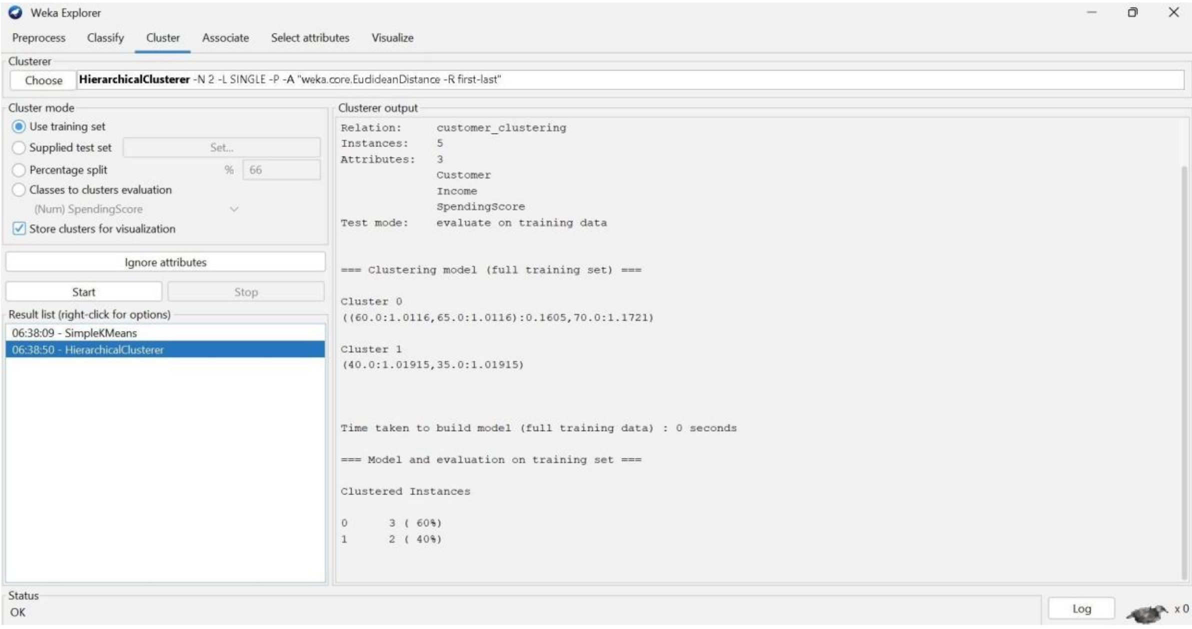
Customer	Income (₹)	Spending Score
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

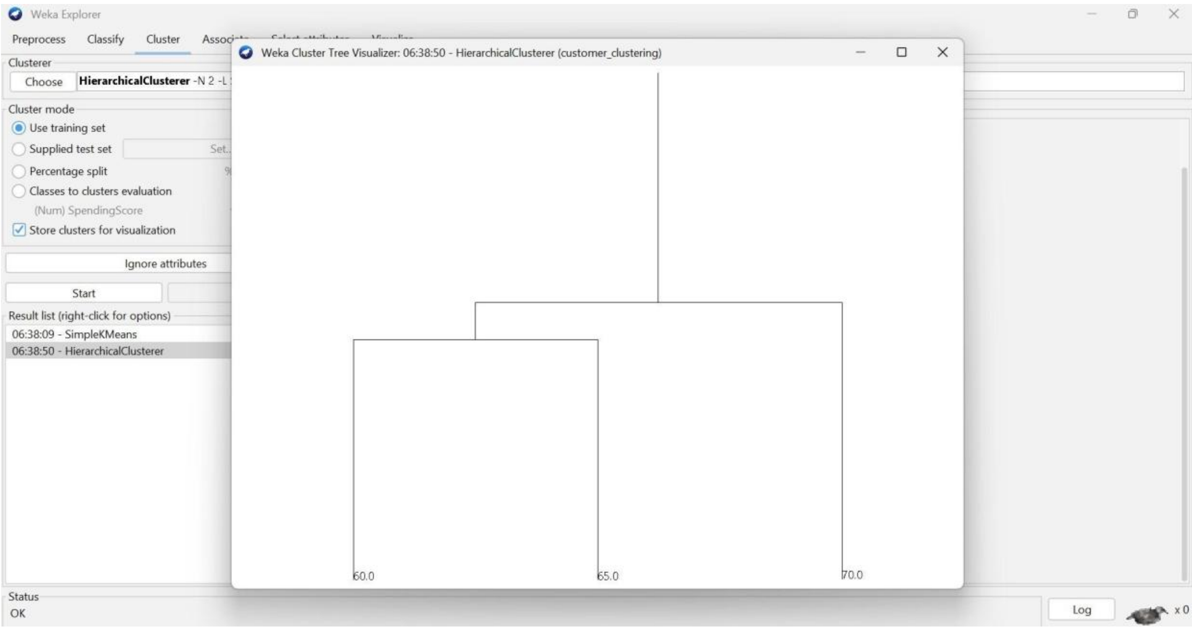
Question:
Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) SpendingScore
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:38:09 - SimpleKMeans
06:38:50 - HierarchicalClusterer
Clusterer output
Cluster 0: C4,28000,65
Cluster 1: C2,60000,40
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (3.0) (2.0)
Customer C1 C1 C2
Income 46600 36000 62500
SpendingScore 54 65 37.5
Time taken to build model (full training data) : 0.02 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 3 (60%)
1 2 (40%)
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Clusterer
Choose
HierarchicalClusterer -N 2 -L
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) SpendingScore
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:38:09 - SimpleKMeans
06:38:50 - HierarchicalClusterer
Weka Clusterer Visualize: 06:38:09 - SimpleKMeans (customer_clustering)
X: Instance_number (Num)
Y: Customer (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: customer_clustering_clustered
C5
C4
C3
C2
C1
0
2
4
X
Y
Class colour
cluster0 cluster1
Status
OK
Log
x 0





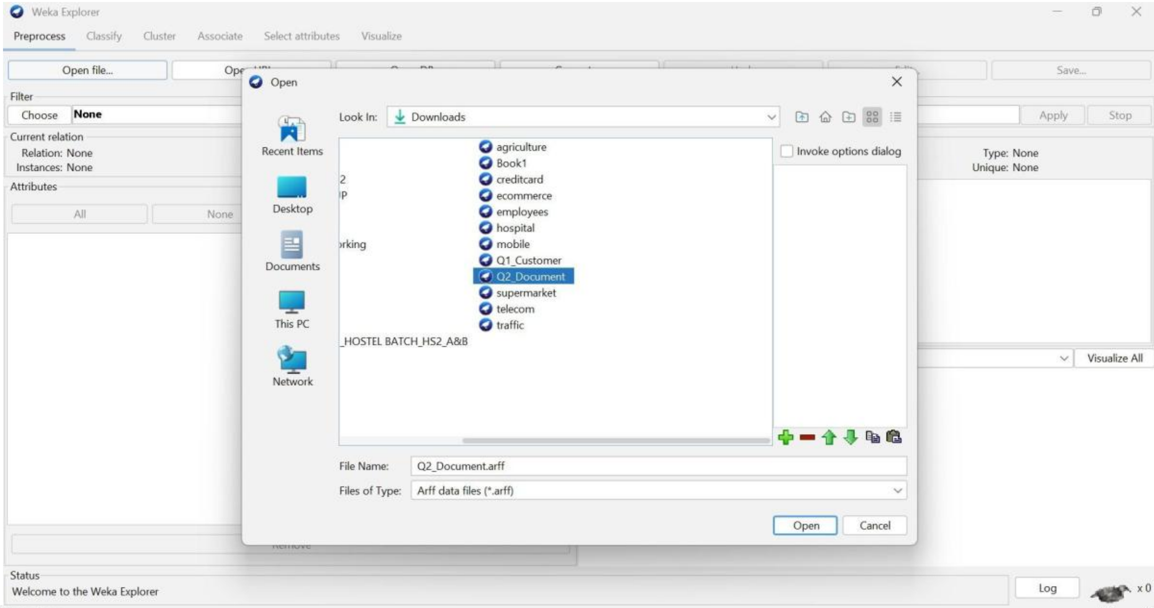
Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

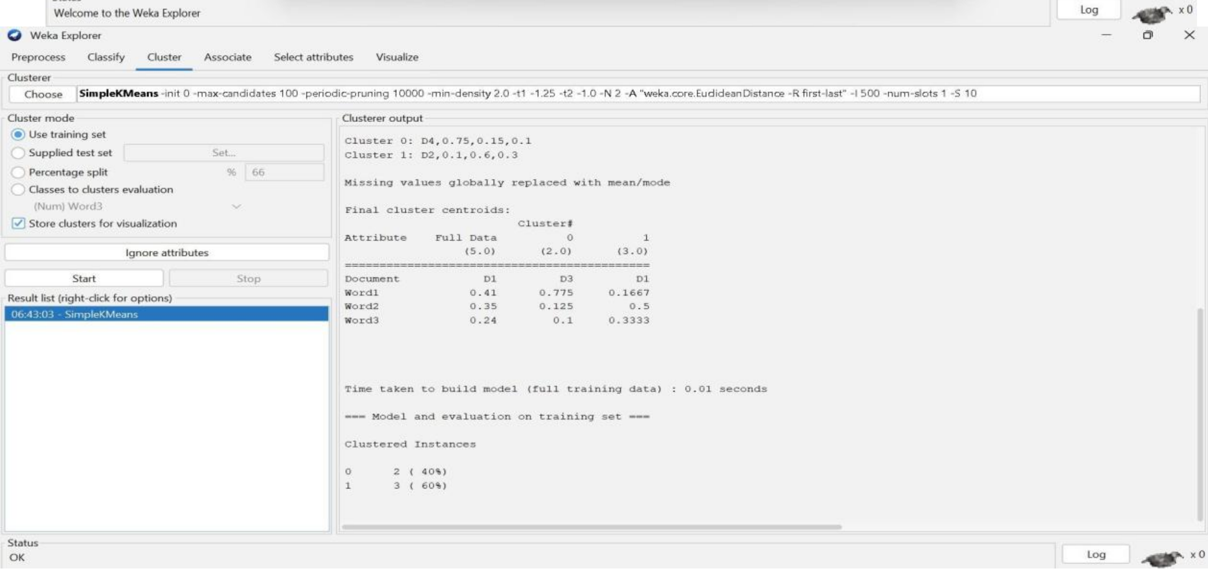
Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark)

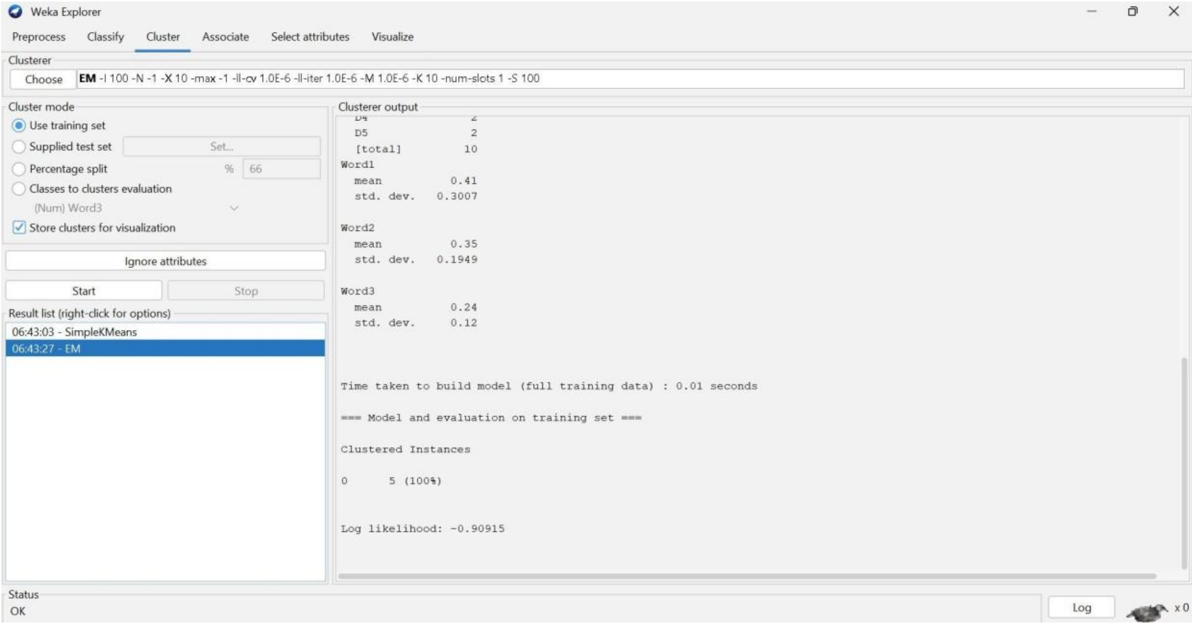
OUTPUT:



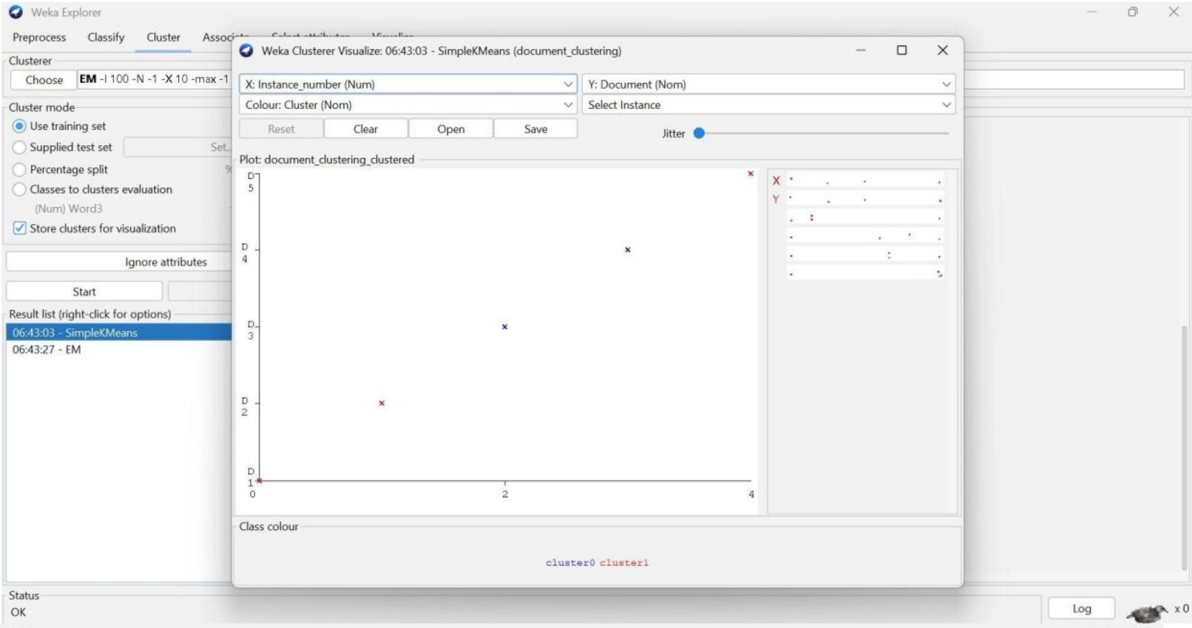
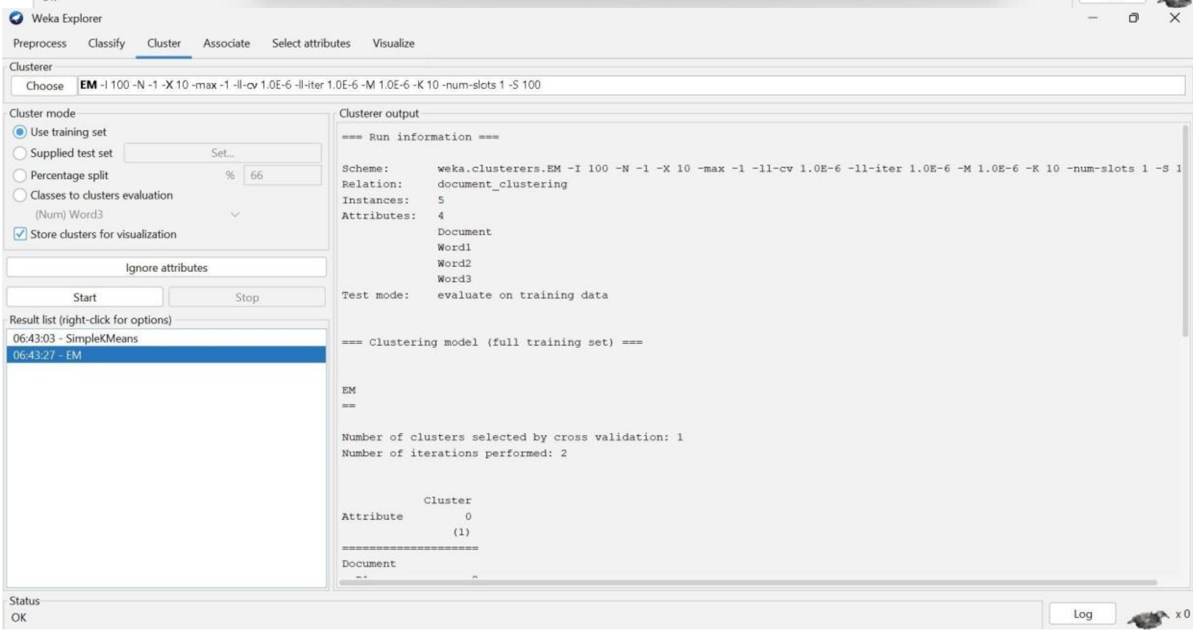
Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open
Look In: Downloads
agriculture
Book1
creditcard
ecommerce
employees
hospital
mobile
Q1_Customer
Q2_Document
supermarket
telecom
traffic
Invoke options dialog
File Name: Q2_Document.arff
Files of Type: Arff data files (*.arff)
Open
Cancel
Status
Welcome to the Weka Explorer

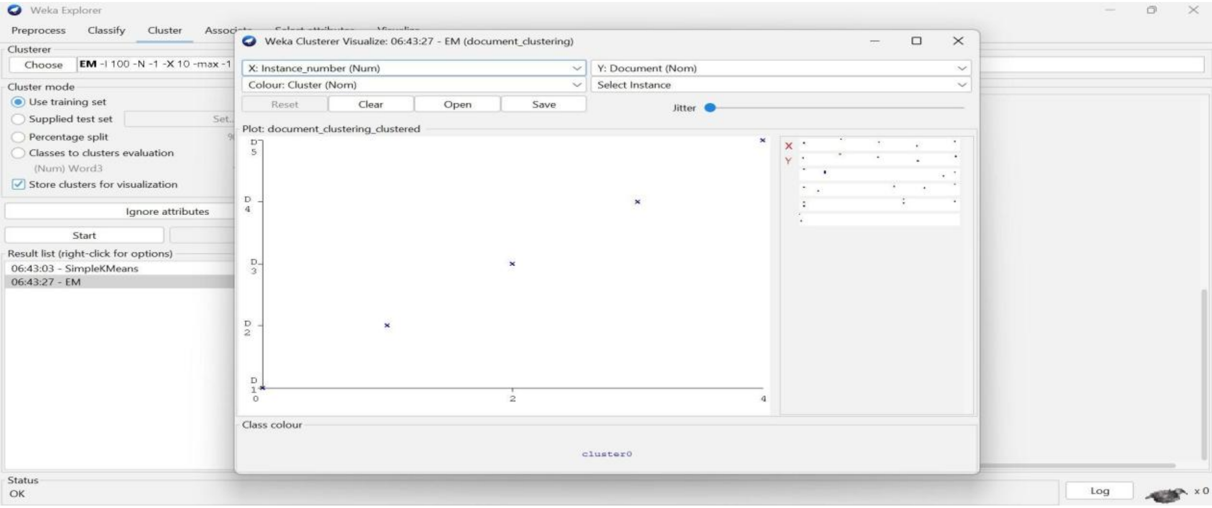


Weka Explorer
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:43:03 - SimpleKMeans
Clusterer output
Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3
Missing values globally replaced with mean/mode
Final cluster centroids:
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Document D1 D3 D1
Word1 0.41 0.775 0.1667
Word2 0.35 0.125 0.5
Word3 0.24 0.1 0.3333
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Word3
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:43:03 - SimpleKMeans
06:43:27 - EM
Clusterer output
D5 2
[total] 10
Word1
mean 0.41
std. dev. 0.3007
Word2
mean 0.35
std. dev. 0.1949
Word3
mean 0.24
std. dev. 0.12
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 5 (100%)
Log likelihood: -0.90915
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Word3
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:43:03 - SimpleKMeans
06:43:27 - EM
Status
OK
Log
x 0
Weka Clusterer Visualize: 06:43:03 - SimpleKMeans (document_clustering)
X: Instance_number (Num)
Y: Document (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: document_clustering_clustered
Class colour
cluster0 cluster1

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Word3
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:43:03 - SimpleKMeans
06:43:27 - EM
Clusterer output
=== Run information ===
Scheme: weka.clusterers.EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 1
Relation: document_clustering
Instances: 5
Attributes: 4
Document
Word1
Word2
Word3
Test mode: evaluate on training data
=== Clustering model (full training set) ===
EM
==
Number of clusters selected by cross validation: 1
Number of iterations performed: 2
Cluster
Attribute 0
(1)
====================
Document
Status
OK
Log
x 0

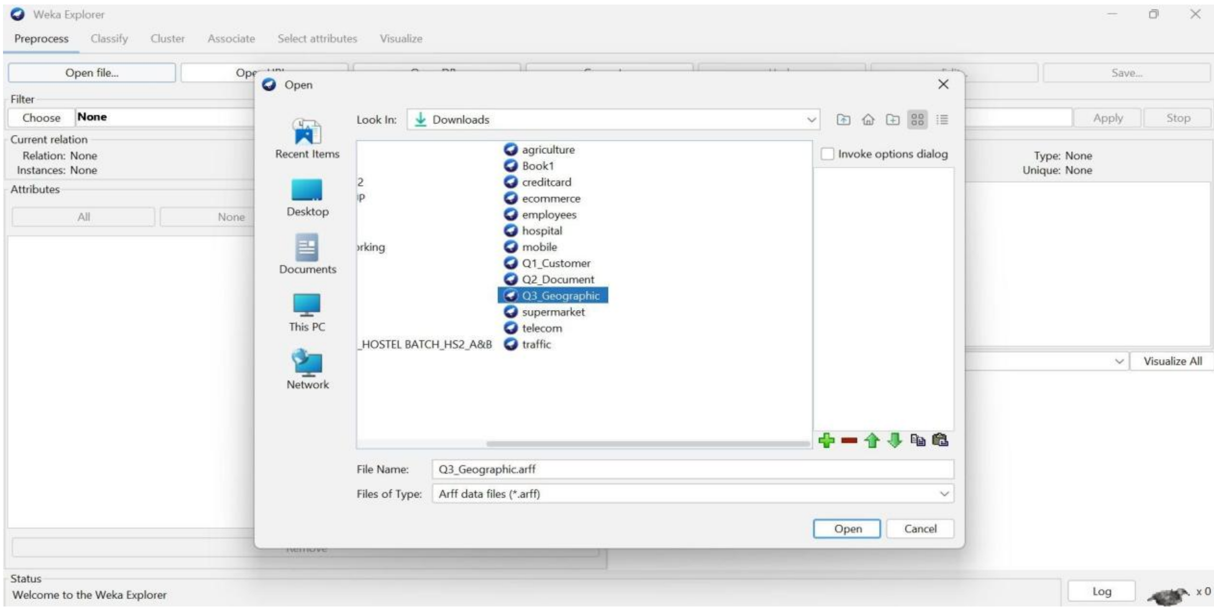


Q3. Dataset: Geographic Data

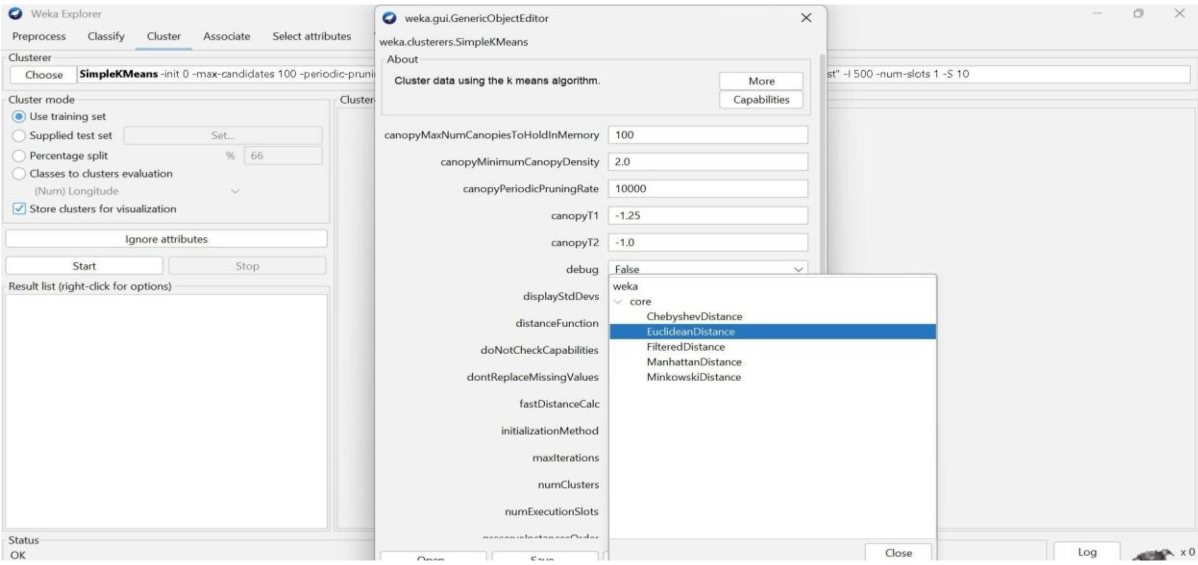
Point	**Latitude**	**Longitude**
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

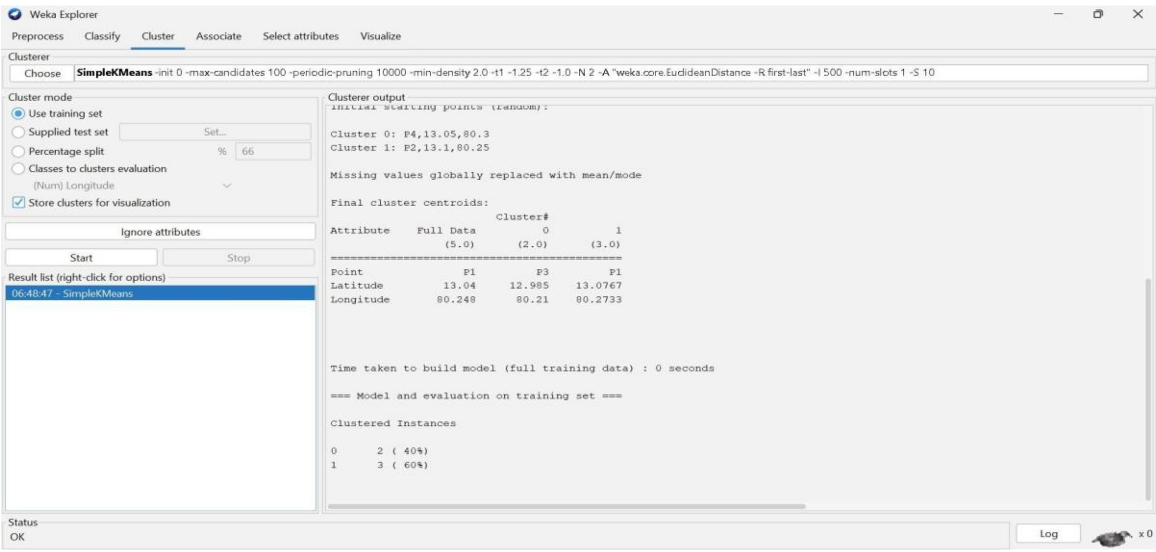
Question:

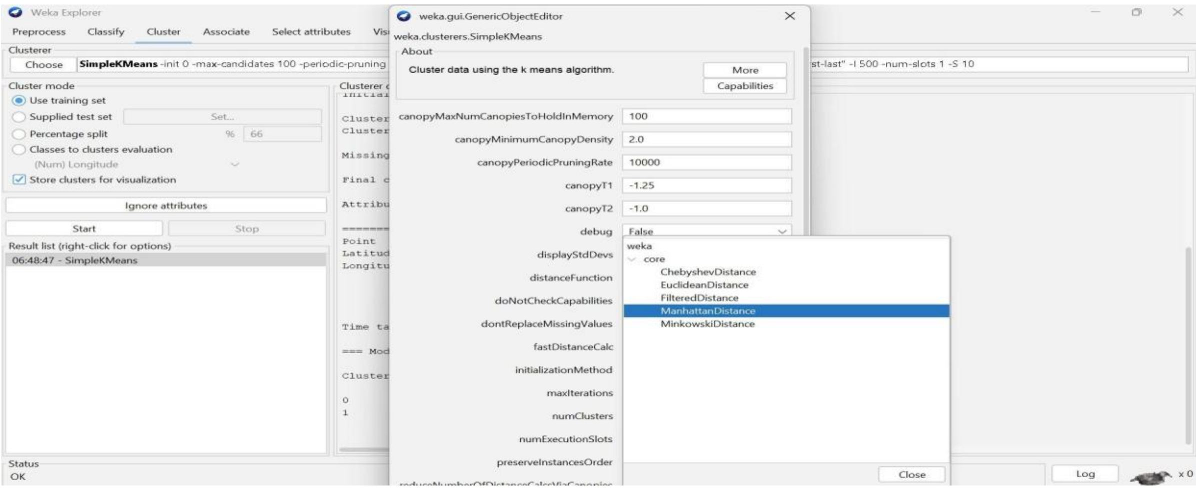
Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

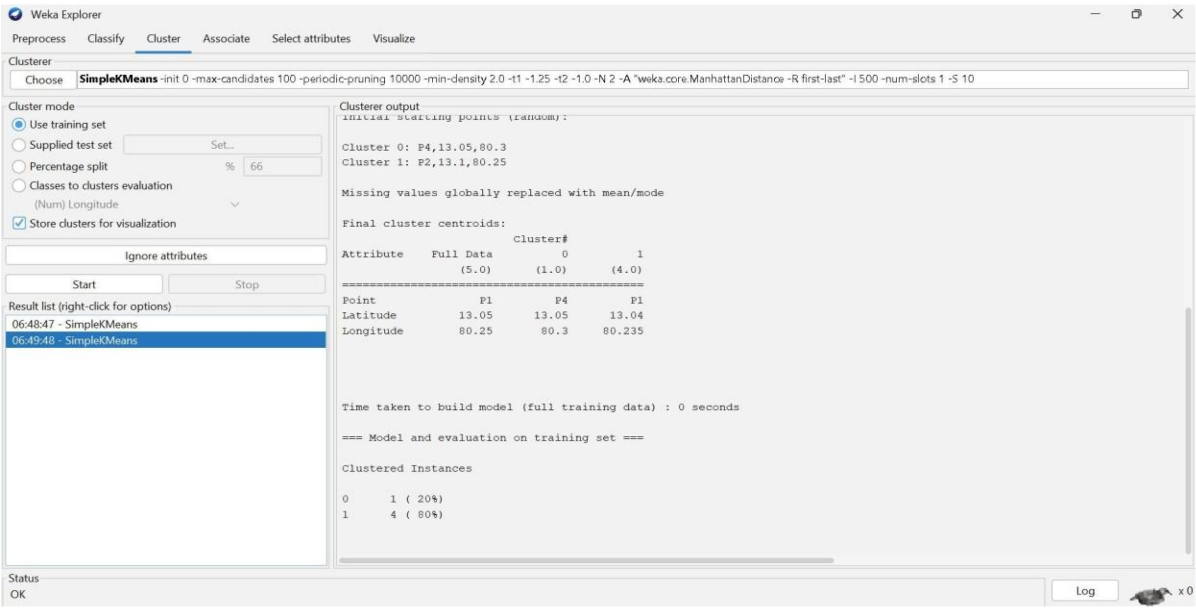


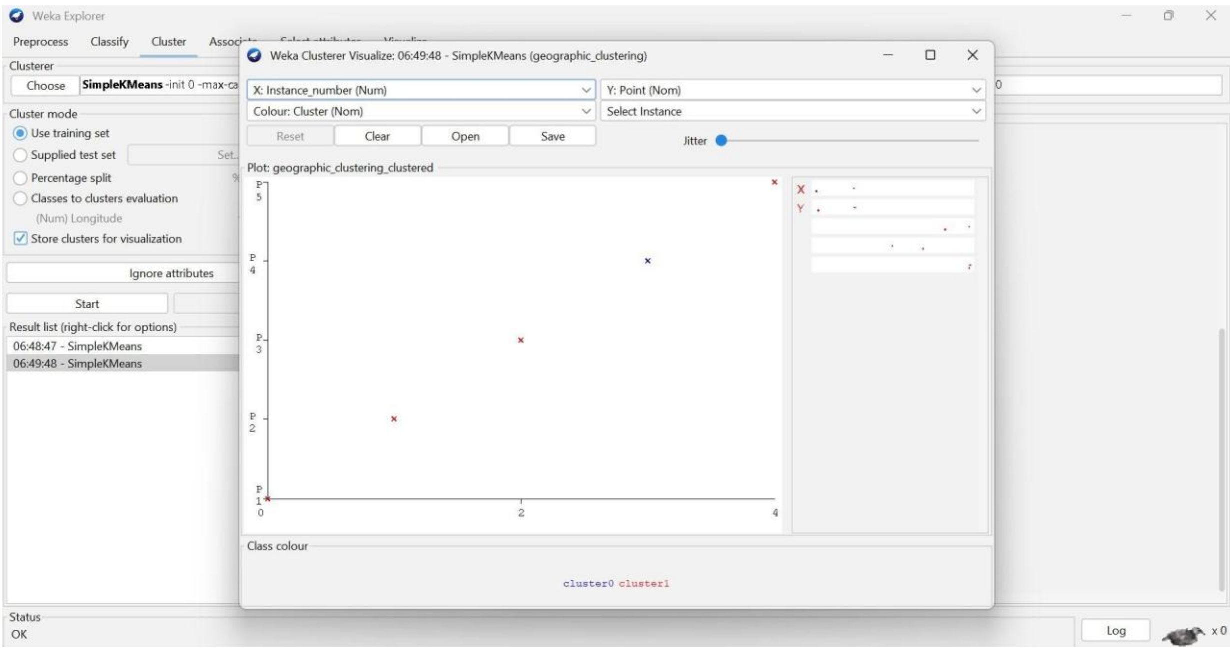
OUTPUT:


Weka Explorer
Preprocess Classify Cluster Associate Select attributes
Clusterer
Choose SimpleKMeans -init 0 -max-candidates 100 -periodic-pruni
st" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Longitude
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
Status
OK
Log
x 0
weka.gui.GenericObjectEditor
weka.clusterers.SimpleKMeans
About
Cluster data using the k means algorithm.
More
Capabilities
canopyMaxNumCanopiesToHoldInMemory 100
canopyMinimumCanopyDensity 2.0
canopyPeriodicPruningRate 10000
canopyT1 -1.25
canopyT2 -1.0
debug False
displayStdDevs
distanceFunction
doNotCheckCapabilities
dontReplaceMissingValues
fastDistanceCalc
initializationMethod
maxIterations
numClusters
numExecutionSlots
weka
core
ChebyshevDistance
EuclideanDistance
FilteredDistance
ManhattanDistance
MinkowskiDistance
Close


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Longitude
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:48:47 - SimpleKMeans
Clusterer output
Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Point P1 P3 P1
Latitude 13.04 12.985 13.0767
Longitude 80.248 80.21 80.2733
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log
x 0


Weka Explorer
Preprocess Classify Cluster Associate Select attributes
weka.gui.GenericObjectEditor
weka.clusterers.SimpleKMeans
About
Cluster data using the k means algorithm.
More
Capabilities
canopyMaxNumCanopiesToHoldInMemory 100
canopyMinimumCanopyDensity 2.0
canopyPeriodicPruningRate 10000
canopyT1 -1.25
canopyT2 -1.0
debug False
displayStdDevs
distanceFunction
doNotCheckCapabilities
dontReplaceMissingValues
fastDistanceCalc
initializationMethod
maxIterations
numClusters
numExecutionSlots
preserveInstancesOrder
weka
core
ChebyshevDistance
EuclideanDistance
FilteredDistance
ManhattanDistance
MinkowskiDistance
Close
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Longitude
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:48:47 - SimpleKMeans
Status
OK
Log


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.ManhattanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Longitude
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:48:47 - SimpleKMeans
06:49:48 - SimpleKMeans
Clusterer output
Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (1.0) (4.0)
Point P1 P4 P1
Latitude 13.05 13.05 13.04
Longitude 80.25 80.3 80.235
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 1 (20%)
1 4 (80%)
Status
OK
Log

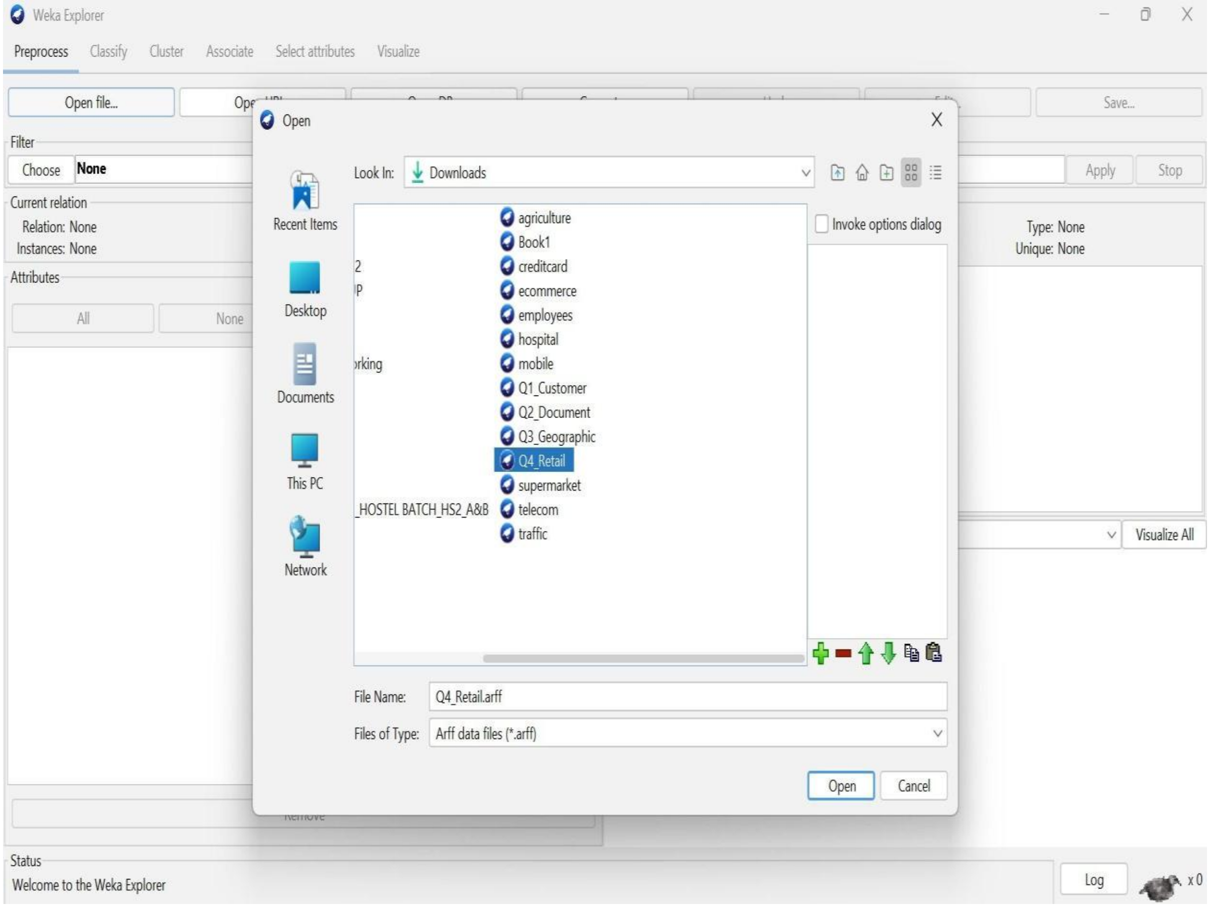

Weka Explorer
Weka Clusterer Visualize: 06:49:48 - SimpleKMeans (geographic_clustering)
X: Instance_number (Num)
Y: Point (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: geographic_clustering_clustered
P5
P4
P3
P2
P1
0
2
4
Class colour
cluster0 cluster1
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Longitude
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:48:47 - SimpleKMeans
06:49:48 - SimpleKMeans
Status
OK
Log

Q4. Dataset: Retail Products

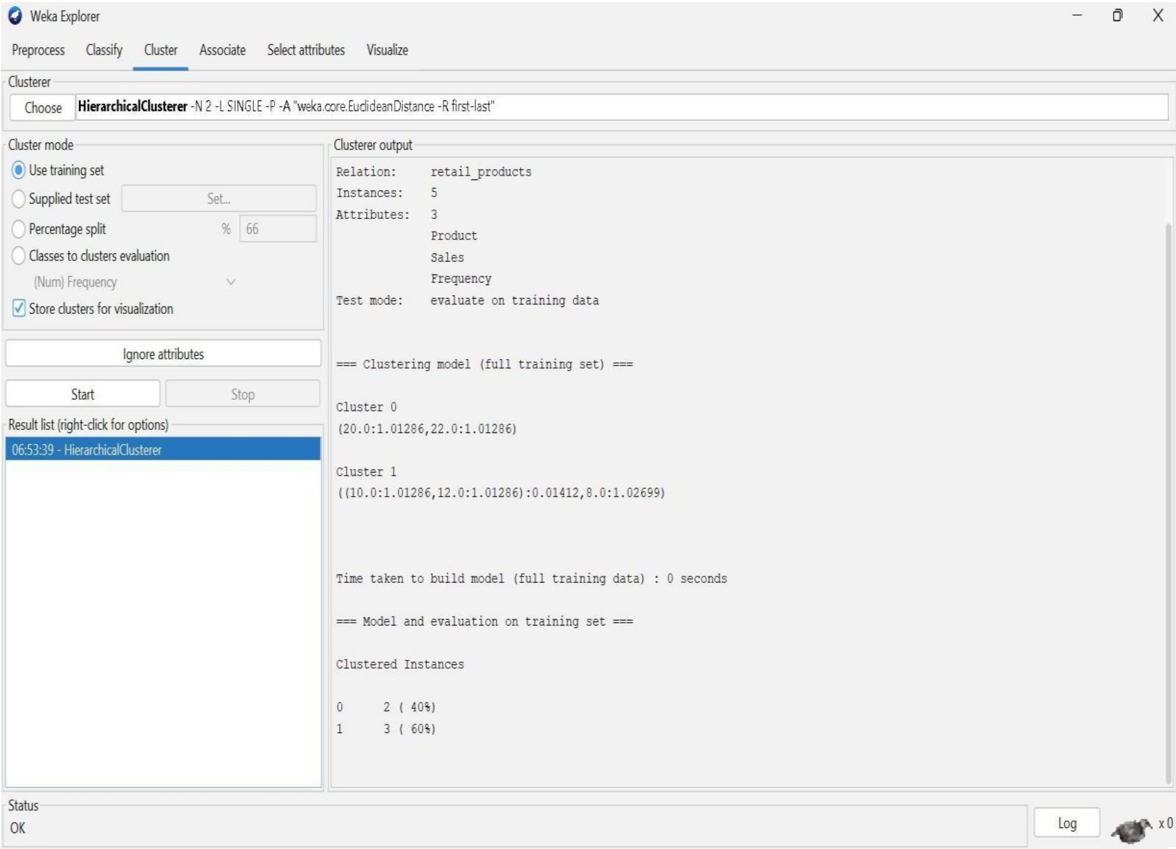
Product	Sales (₹)	Frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

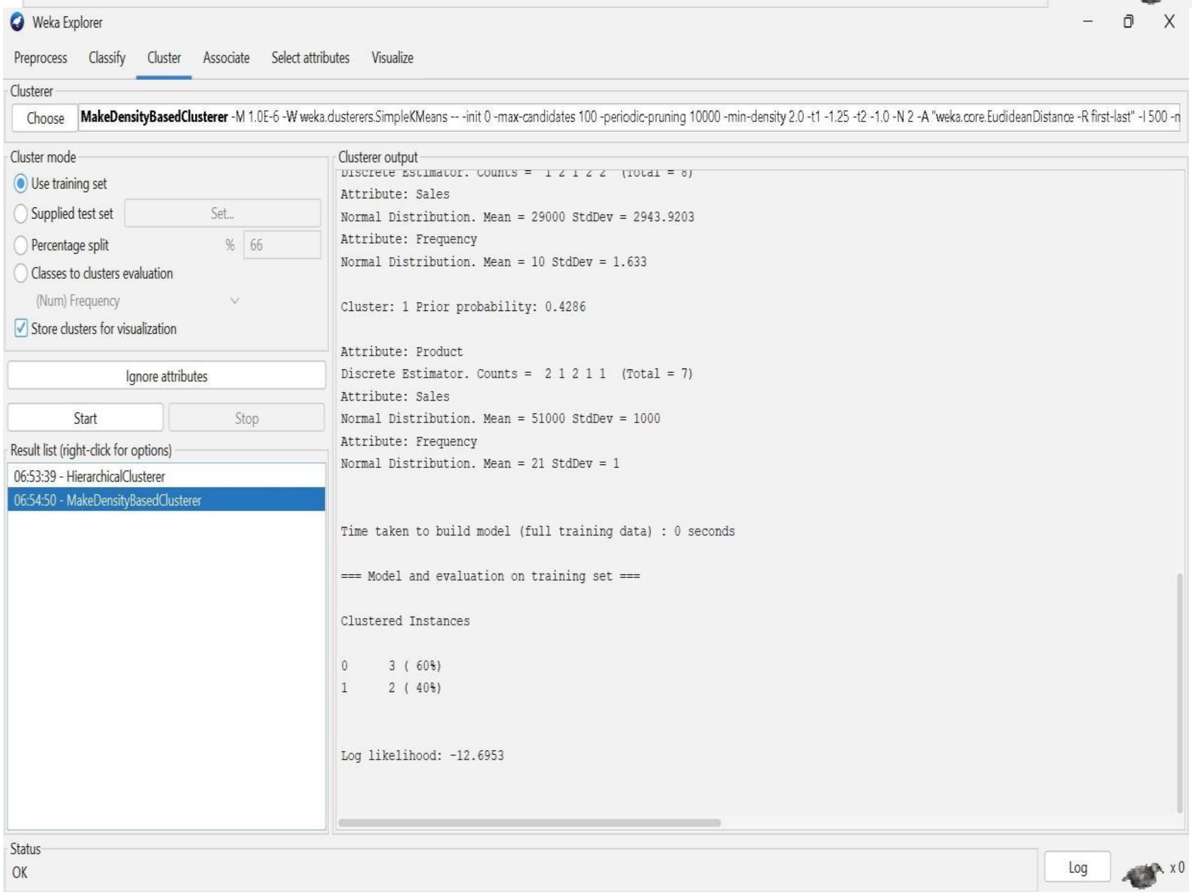
Question:

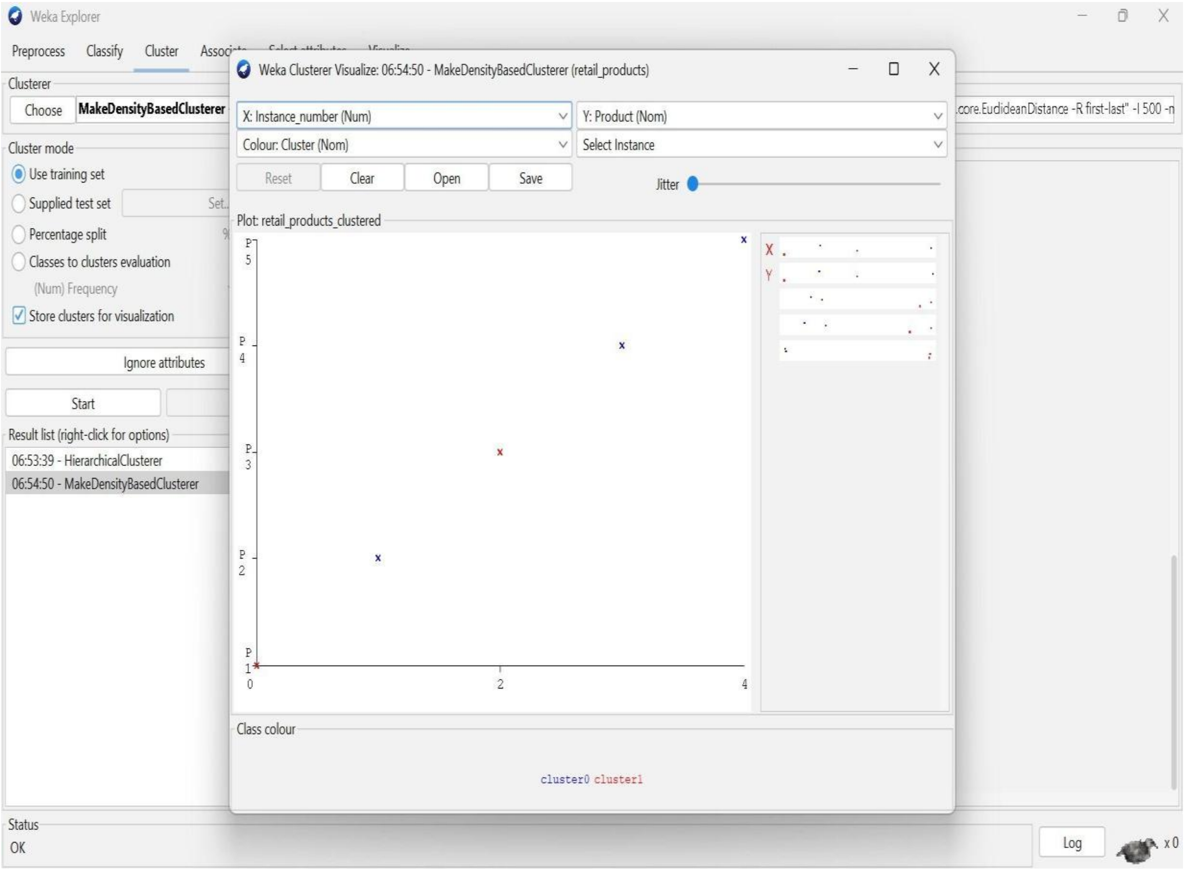
Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*



OUTPUT:


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Frequency
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:53:39 - HierarchicalClusterer
Clusterer output
Relation: retail_products
Instances: 5
Attributes: 3
Product
Sales
Frequency
Test mode: evaluate on training data
=== Clustering model (full training set) ===
Cluster 0
(20.0:1.01286,22.0:1.01286)
Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log
x 0


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose MakeDensityBasedClusterer -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Frequency
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:53:39 - HierarchicalClusterer
06:54:50 - MakeDensityBasedClusterer
Clusterer output
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633
Cluster: 1 Prior probability: 0.4286
Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 3 (60%)
1 2 (40%)
Log likelihood: -12.6953
Status
OK
Log
x 0



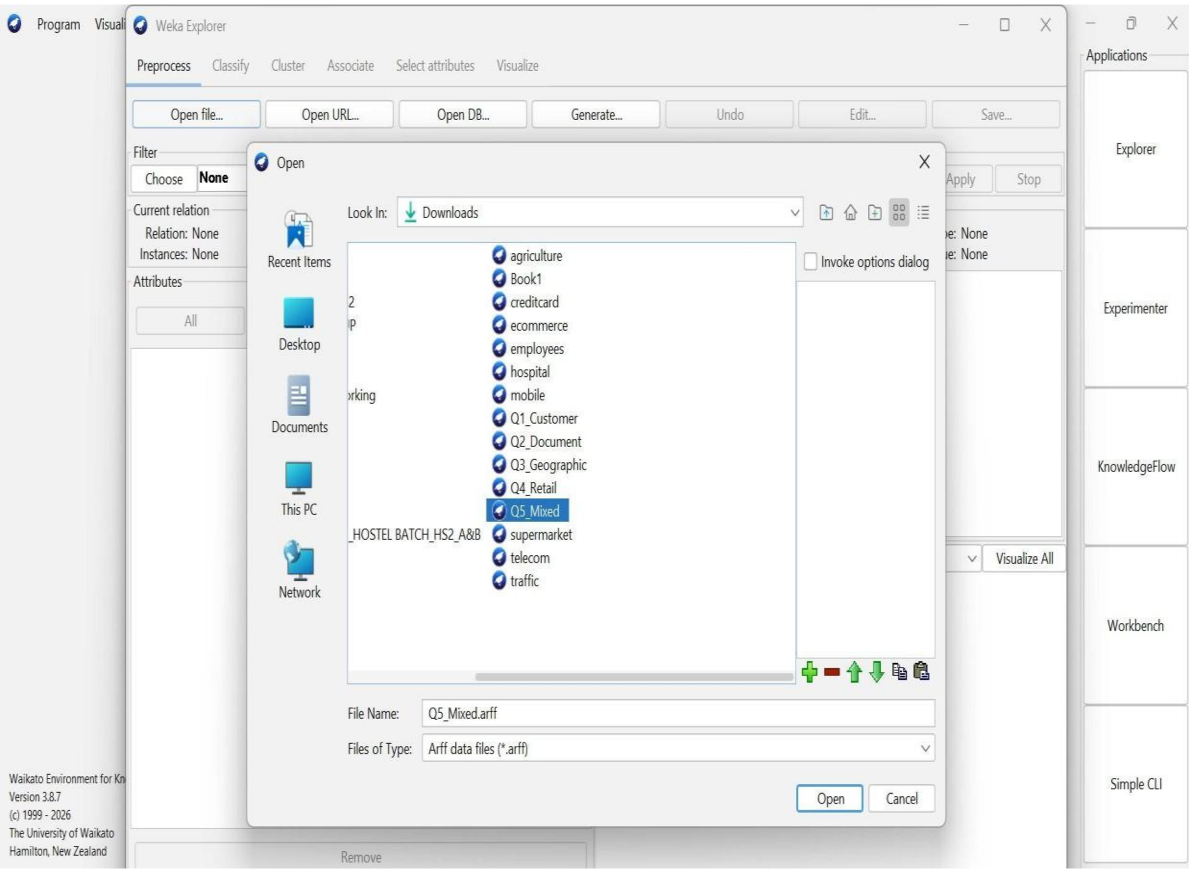
Weka Explorer
Preprocess
Classify
Cluster
Clusterer
Choose
MakeDensityBasedClusterer
core.EuclideanDistance -R first-last" -I 500 -n
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Frequency
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:53:39 - HierarchicalClusterer
06:54:50 - MakeDensityBasedClusterer
Status
OK
Log
x 0
Weka Clusterer Visualize: 06:54:50 - MakeDensityBasedClusterer (retail_products)
X: Instance_number (Num)
Y: Product (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: retail_products_clustered
Class colour
cluster0 cluster1

Q5 . Dataset: Mixed Attributes

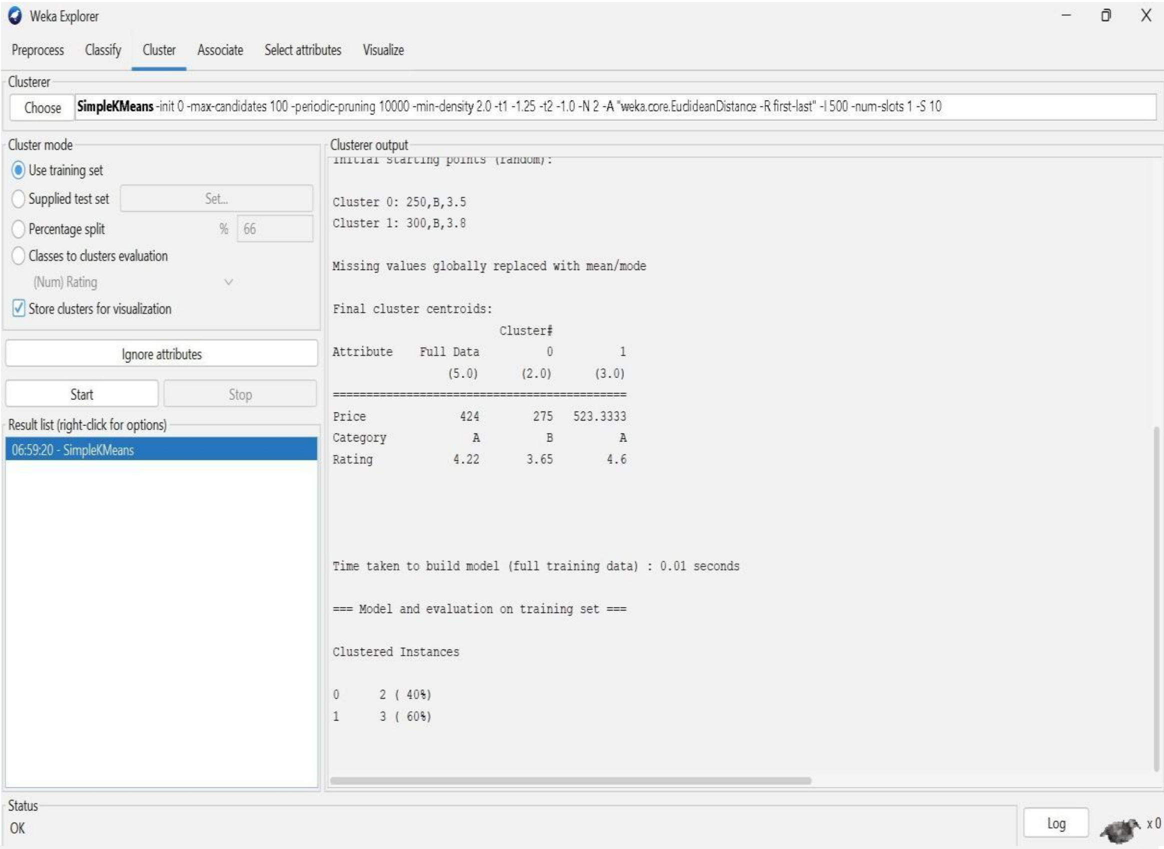
Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

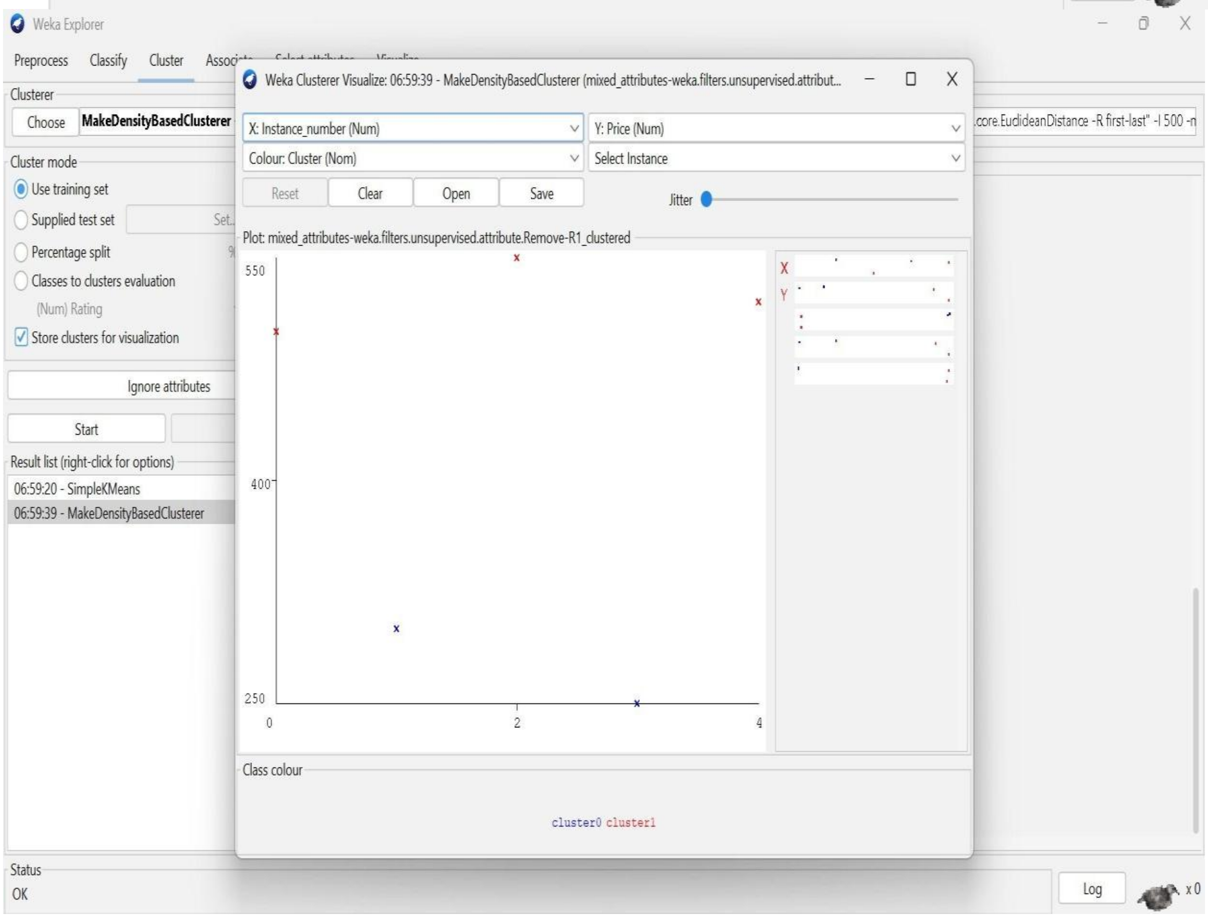
Question:

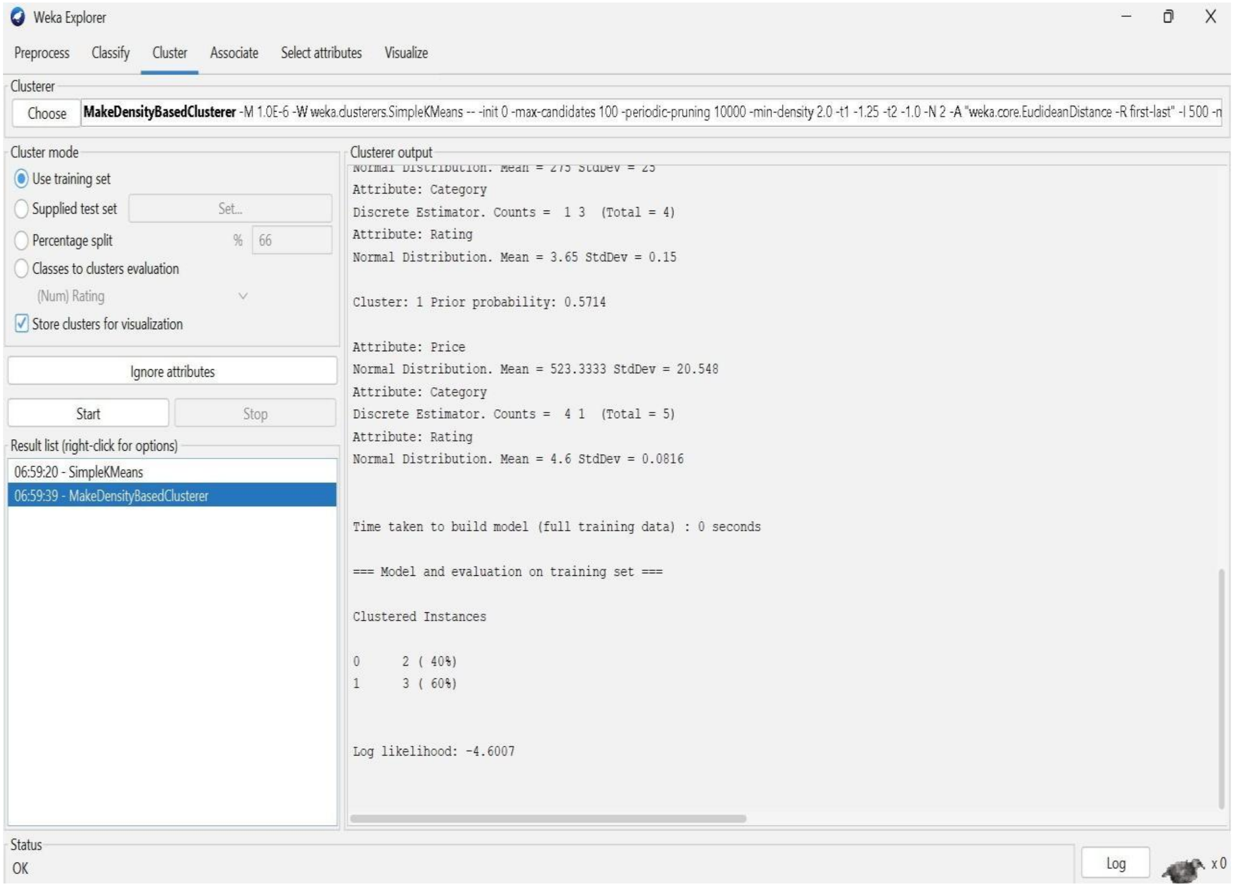
Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*



OUTPUT:


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Rating
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:59:20 - SimpleKMeans
Clusterer output
Cluster 0: 250,B,3.5
Cluster 1: 300,B,3.8
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Price 424 275 523.3333
Category A B A
Rating 4.22 3.65 4.6
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Clusterer
Choose
MakeDensityBasedClusterer
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Rating
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:59:20 - SimpleKMeans
06:59:39 - MakeDensityBasedClusterer
Weka Clusterer Visualize: 06:59:39 - MakeDensityBasedClusterer (mixed_attributes-weka.filters.unsupervised.attribut...
X: Instance_number (Num)
Y: Price (Num)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: mixed_attributes-weka.filters.unsupervised.attribute.Remove-R1_clustered
550
400
250
0
2
4
Class colour
cluster0 cluster1
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
MakeDensityBasedClusterer -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Rating
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:59:20 - SimpleKMeans
06:59:39 - MakeDensityBasedClusterer
Clusterer output
Attribute: Category
Discrete Estimator. Counts = 1 3 (Total = 4)
Attribute: Rating
Normal Distribution. Mean = 3.65 StdDev = 0.15
Cluster: 1 Prior probability: 0.5714
Attribute: Price
Normal Distribution. Mean = 523.3333 StdDev = 20.548
Attribute: Category
Discrete Estimator. Counts = 4 1 (Total = 5)
Attribute: Rating
Normal Distribution. Mean = 4.6 StdDev = 0.0816
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Log likelihood: -4.6007
Status
OK
Log
x 0